PART 2 Instructions

Now, we explain how your payoff will be determined.

The bottom line is that **the more accurate your estimate is, the more likely you will earn a higher payoff.**

How is your payoff determined?

Step 1: At the end of Part 2, your penalty score for each of your estimates will be calculated in the following way. *You will receive a higher penalty score from Box A (or from Box B) as your estimate deviates more from the actual number of Green balls you (or your teammate) received.*

$$Penalty\ Score\ from\ Box\ A = \sum_{i=1}^{10} \left(tokens\ in\ pillar\ i - 100 * I_i^A\right)^2$$

$$Penalty\ Score\ from\ Box\ B = \sum_{i=1}^{10} \left(tokens\ in\ pillar\ i - 100 * I_i^B\right)^2$$

where $i = 1, \ldots, 10$ refers to the different bins in each of Box A and Box B. I_i^A equals 1 for the bin in Box A which contains the number of the actual Green balls you received and 0 otherwise. I_i^B equals 1 for the bin in Box B which contains the number of the actual Green balls your teammate received and 0 otherwise.

Step 2: Then, a number X^A is drawn randomly from the interval [0,20000]. Any number within the interval [0,20000] is equally likely to be selected. Independent of X^A, a number X^B is also drawn in the same way. If your $Penalty\ Score\ from\ Box\ A$ is less than or equal to the random number X^A, you will earn 30 Experimental Dollars. Otherwise, you will zero. If your $Penalty\ Score\ from\ Box\ B$ is less than or equal to the random number X^B, you will earn 30 Experimental Dollars. Otherwise, you will earn zero. *In other words, the higher penalty score you get, the lower probability you earn 30 Experimental Dollars for each of your estimates.*

To help your understanding, suppose you received 55 Green balls in Part 1. Instead of informing you of the number of Green balls you received in the experiment, the experiment requires you to report your estimate by allocating 100 tokens to each of 10 bins in Box A. Notice that, in this example, the bin that contains the number of the actual Green balls you received is *bin "51-60"*.

Consider the following scenarios.

- **Scenario 1:**

  You feel very confident that you received more than (or equal to) 51 but less than (or equal to) 60 Green balls in Part 1. Therefore, you allocated all the 100 tokens to "51-60" bin. If this is the case, you receive no penalty because you estimate is very accurate. You earn 30 Experimental Dollars with certainty (100 percent chance).

Bins	**0-10**	**11-20**	**21-30**	**31-40**	**41-50**	**51-60**	**61-70**	**71-80**	**81-90**	**91-100**	**Penalty Score**
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	0	0	0	0	0	100	0	0	0	0	
	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(100 - 100)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	*0*

- **Scenario 2:**

  You feel very confident that you received more than (or equal to) 41 but less than (or equal to) 50 Green balls in Part 1. Therefore, you allocated all the 100 tokens to "41-50" bin. If this is the case, your penalty score is 20000 (the highest possible penalty score) because your estimate is completely wrong. There is no chance of earning 30 Experimental Dollars (0 percent chance).

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	0	0	0	0	100	0	0	0	0	0	
	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(100 - 0)^2$	$(0 - 100)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	*20000*

- **Scenario 3:**

  You are guessing you received 41-50 Green balls with 50 percent chance and 51-60 Green balls with 50 percent chance. Therefore, you allocated all the 50 tokens to "41-50" bin and the rest of 50 tokens to "51-60" bin. If this is the case, your penalty score is 5000. You earn 30 Experimental Dollars with 75 percent chance.

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	0	0	0	0	50	50	0	0	0	0	
	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(50 - 0)^2$	$(50 - 100)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	*5000*

- **Scenario 4:**

  You are guessing that you received 41-50 Green balls with 30 percent chance and 51-60 Green balls with 70 percent chance. Therefore, you allocated all the 30 tokens to "41-50" bin and the rest of 70 tokens to "51-60" bin. If this is the case, your penalty score is 1800. You earn 30 Experimental Dollars with 91 percent chance.

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	0	0	0	0	30	70	0	0	0	0	
	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(30 - 0)^2$	$(70 - 100)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	$(0 - 0)^2$	*1800*

- **Scenario 5:**

  You decided to just randomly distribute the 100 tokens to each bin. If this is the case, your penalty score is 9000. You earn 30 Experimental Dollars with 55 percent chance.

Bins	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100	Penalty Score
Does this bin contain the actual number of Green balls?	No	No	No	No	No	Yes	No	No	No	No	
# tokens you distributed to this bin	10	10	10	10	10	10	10	10	10	10	
	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 100)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	$(10 - 0)^2$	*9000*

In summary, you earn 30 Experimental Dollars with certainty if your estimate is very accurate. There is no chance you earn 30 Experimental Dollars if your estimate is completely wrong. **To increase the chance of earning 30 points, you should provide an estimate as accurate as possible.**

On the next screen, you are presented with comprehension questions.

You can review the instructions by clicking "Back".

Once you get familiar with Part 2 instructions, click "Next".

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Debug info

Basic info

ID in group	2
Group	43
Round number	1
Participant	P2
Participant label	
Session code	6hblo8hf